

Optimal Time-Consistent Dollar Liquidity Provision

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Abstract

I study the interaction of dollar swap lines with U.S. safe asset market power. The United States can exercise market power over its assets' liquidity premium provided dollar liquidity is credibly scarce during a crisis. Yet when crises occur, it has an ex post incentive to provide abundant dollar liquidity through swap lines to prevent disorderly sales of U.S. debt. In a two-country model of optimal safe asset issuance, I show that this time-consistency problem erodes U.S. market power and increases the price elasticity of foreign demand for Treasuries. I bring this prediction to the data using the swap-line interventions of 2008 and March 2020. Relative to countries excluded from the network, recipients exhibit the predicted increase in the price elasticity of Treasury demand after each episode. Matching these demand rotations in a calibrated structural model implies that the unanticipated component of the interventions reduced the convenience yield to less than half its initial value.

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1 Introduction

Foreign financial firms require access to dollar funding in a crisis. If U.S. liquidity provision is stringent in bad times, foreign banks must hold precautionary buffers of safe and liquid U.S. assets. Buffer demand is inelastic, the United States is a monopoly supplier, and it prices accordingly. A markup, or a “convenience yield,” prevails on U.S. public debt. But in a crisis, those banks liquidate their buffers, depressing the price of the debt precisely when the United States’ need to smooth consumption is greatest. The United States therefore prefers liquid foreign markets and eases funding stress in bad times — a policy chosen *today*, only after *yesterday’s* buffers were already sold. Foreign banks anticipate relief, flattening the pre-crisis demand curve the United States faces. The inability to commit to non-provision of crisis liquidity erodes U.S. monopoly power and depresses the convenience yield.

Demand for U.S. Treasuries is notably inelastic, and Choi et al. (2026) provide compelling evidence that the United States internalizes the market power this implies when issuing debt. Yet Jiang et al. (2025) and Du et al. (2026) measure large declines in the convenience yield following the Global Financial Crisis (GFC) and the onset of COVID-19. What caused the United States to lower its premium? This paper proposes the above-stated dynamic inconsistency as an explanation for the observed decline in the convenience yield. Each crisis featured extensive — and possibly unanticipated — U.S. liquidity support through dollar swap lines. Even at the monopolistic optimal policy, innovations to swap line provision cause the United States to erode its own market power and lower the convenience yield.

I characterize optimal time-consistent policy in a two-country setting. The United States chooses debt and swap line crisis support subject to a reduced-form institutional constraint. Liquid asset demand is inelastic because foreign banks experience liquidity needs in crisis. I first show formally that swap line access flattens the demand curve for U.S. debt, inducing a pointwise increase in the demand elasticity under minimal assumptions. Intuitively, swap lines place a ceiling on dollar funding spreads, *bounding* the level but *eliminating* the slope of assets’ liquidity value. Second, I show whenever U.S. borrowing and Foreign liquidity needs co-occur, the time-consistent policy provides the maximum swap line support permitted by the constraint. Relaxing that constraint translates directly into greater liquidity provision,

a flatter demand curve, and weaker U.S. market power.

These theoretical results provide the key to empirical discipline. Swap lines are distinctive in moving the slope, not just the level, of demand. I estimate the difference-in-differences of the out-of-crisis elasticity of demand between swap line recipients and non-recipients. The design recovers a *rotation* in the demand curve, rather than a shift of it, acting where the theoretical prediction is cleanest and leaving unrestricted the ambiguous effect of swap line access on the level of demand. *Relative* demand curve rotations purge the endogenous response of policy, which confounds the aggregate elasticity under monopoly pricing. Estimating on *non-crisis* buffer-stock accumulation gives sufficient observations of reliable annual cross-country holdings data. Following both the 2008 and 2020 expansions of swap line access, I reject the null of no increase in recipients' elasticity of demand.

Swap lines were of course not randomly assigned, and while I control for observable sources of selection I do not interpret these results as causal. The design nevertheless places demanding restrictions on potential confounds. Baseline elasticities are differenced out: a competing explanation must not merely predict that recipients have flatter demand, but that their demand became flatter at the timing of each episode. I focus on the most plausible sources of such selection, including differences in Treasury holdings and financial development. These characteristics primarily predict levels of demand and, insofar as they also predict changes in its slope, work against the estimated rotation. Controlling for them increases the total point estimate and strengthens the rejection of no rotation. The estimates constitute a suggestive agreement of theory with data, and provide a clean empirical target for structural inference.

The time-consistent policy is not simply static monopoly pricing, and the change in elasticity is not a sufficient statistic for the change in the premium. To recover the latter I solve and calibrate a structural version of the dynamic two-country model. Replicating the estimated demand rotations implies a reduction in convenience yields of over half their initial value resulting from the unanticipated part of the swap line policy. Expansion of emergency dollar liquidity facilities in a time-consistent world is a theoretically plausible and quantitatively powerful explanation of falling convenience yields on U.S. public debt.

1.1 Related Literature

The importance of safe and liquid dollar-denominated assets to the international financial system is long established, with Coppola et al. (2023) and Gopinath and Stein (2021) providing modern treatments. The idea that a large asset issuer may exert market power over such assets was given theoretical grounding by Farhi and Maggiori (2018) and Costinot et al. (2014).

Choi et al. (2026) provide evidence that such market power is exerted in practice, combining empirical tools from industrial organization with recent framings of inelastic asset demand.

These authors draw on a large literature studying the convenience yield on U.S. Treasuries. Krishnamurthy and Vissing-Jorgensen (2012) provide foundational estimates of a premium accruing to U.S. public debt among domestic investors, while Du et al. (2018), Jiang et al. (2021), and Kojen and Yogo (2026) extend this to infer larger premia paid by international investors. A range of papers consider the fiscal and macroeconomic implications of these spreads (Kekre and Lenel, 2024; Jiang et al., 2024; Devereux et al., 2026; Bodenstein et al., 2025; Choi et al., 2024). Evidence of a secular decline in the convenience yield in the post-crisis period is provided by Du et al. (2026) and Jiang et al. (2025), who demonstrate the sensitivity of these premia to issuance shocks.

Evidence of precautionary buffer-stock accumulation of U.S. public debt by foreign banks is provided by Bianchi et al. (2025) and supported by the empirical findings of Chaudhary et al. (2026).

A large literature provides causal evidence of the effects of international dollar liquidity policies on liquidity spreads, bank lending, asset prices, and exchange rates. Many studies focus on high-frequency identification from swap-line announcements (Bahaj and Reis, 2022; Kekre and Lenel, 2025; Aizenman et al., 2022) and QE surprises (Lee and Engel, 2025). Obstfeld et al. (2009) provide a canonical discussion of swap-line and reserve accumulation substitutability.

Beyond the dollar-Treasury nexus, much recent work addresses the wider econometrics of a quantity-based demand-system approach to asset pricing (Kojen and Yogo, 2019; Kojen

and Yogo, 2026; Koijen and Yogo, 2025; Gabaix and Koijen, 2023). Choi et al. (2026) and Moretti et al. (2025) provide the key precursors to this paper in bringing such a framework to bear on policy design, with the latter sharing this paper’s emphasis on limited commitment.

1.2 Structure of the Paper

The two-country environment and the U.S. government debt-issuance problem are introduced in section 2. The key theoretical results are developed in section 3. Informed by these results, section 4 estimates the between-country change in elasticities following the GFC and March 2020. Taking these estimates as targeted moments, section 5 recovers the resulting fall in the convenience yield in a structural model. Section 6 concludes.

2 Liquidity Supply and Inelastic Demand

This section introduces a model of the supply and demand of liquid assets. There are two countries, Foreign and Domestic, with Domestic variables indexed $*$. Countries trade a homogeneous global consumption good and a single non-contingent zero-coupon liquid bond $b_t \geq 0$ which matures geometrically at rate μ . Time is discrete, and a period t splits into “morning” t_- and “afternoon” t_+ . A payment received at t_- is called liquid.

The Foreign country contains banks, which face liquidity shocks, and households.

The Domestic policymaker implements an optimal time-consistent issuance strategy. The policymaker may also provide within-period liquidity support to Foreign banks.

2.1 Foreign Demand

The Foreign country contains a continuum of competitive, profit-maximizing banks and a unit mass of risk-neutral households. Households receive an endowment which banks may take as deposits and profitably invest. Households own the banks and receive their profits as dividends. In crisis episodes, households may demand early repayment of deposits, inducing a resource cost for banks.

2.1.1 Households

A household i receives at the beginning of period t a total endowment $\delta_{i,t}$. They choose deposits $d_{i,t} \leq \delta_{i,t}$. Deposits withdrawn at t_+ accrue a gross return R_t^d ; deposits may be

withdrawn at t_- with zero net return. Endowment that is not deposited may be stored at gross rate $s < 1$. Households hold equal shares in the banks, whose profits $\Pi_{i,t}$ are paid out at t_+ and which they take as given.

At t_- , after the deposit decision has been made, each household learns whether it is patient or impatient. A patient household derives utility from consumption at t_+ , an impatient household from consumption at t_- . Since deposits can be redeemed at par in the morning, they weakly dominate storage, and $d_{i,t} = \delta_{i,t}$ (Appendix B gives the full argument). The deposit rate is an indeterminate transfer within the Foreign country and is immaterial to prices and allocations, so I set $R_t^d = 1$ without loss.

Denote by λ_t the fraction of households who are impatient at t , and take $\beta \in (0, 1)$. Given linear utility the households of a bank can be trivially aggregated, and solve

$$\begin{aligned} \max_{\{c_{i,t}, d_{i,t}\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t c_{i,t} \\ \text{s.t.} \quad & c_{i,t} = [R_t^d - \lambda_t(R_t^d - 1)] d_{i,t} + \Pi_{i,t}. \end{aligned} \tag{1}$$

Scale is inconsequential for the theory, so for exposition I normalize $\delta_{i,t} = 1$, discussing the general case in Appendix B.

2.1.2 Banks

A representative bank enters period t with liquid asset stock b_{t-1} . It takes deposits d_t and invests them in an illiquid within-period technology, which returns Ad_t at t_+ , where A is the technology's gross return. The investment can be unwound early, but only at a cost: obtaining x_t units of morning goods requires forgoing $\Psi(x_t)$ units of afternoon goods.

At t_-

In the morning, the bank learns λ_t . A portion μ of its liquid assets mature. Trade then takes place in the liquid asset. New and previously-issued bonds are perfect substitutes and trade at a common price p_t , so only the bank's net position is relevant.

The bank's liquidity needs consist of early repayment $\lambda_t d_t$, and the value of assets it wishes to purchase, $p_t b_t$. It meets these liquidity needs through maturing assets μb_{t-1} and sales of liquid assets $p_t(1 - \mu)b_{t-1}$. It may also receive within-period liquidity support ℓ_t from the Domestic policymaker, discussed below. Any remaining liquidity shortfall requires early

liquidation of investment x_t . Support is extended at t_- and repaid at par at t_+ : it is a pure transfer of goods across the two halves of the period.

The bank's morning budget constraint is

$$\lambda_t d_t + p_t (b_t - (1 - \mu)b_{t-1}) = \mu b_{t-1} + x_t + \ell_t. \quad (2)$$

At t_+

In the afternoon, the bank's investments are realized with value Ad_t net of early liquidation $\Psi(x_t)$. The bank repays remaining deposits $(1 - \lambda_t)R_t^d d_t$ and any intra-period loan ℓ_t . It rebates the surplus as profits Π_t to households. Its afternoon budget constraint is

$$\Pi_t + (1 - \lambda_t)R_t^d d_t = Ad_t - \Psi(x_t) - \ell_t. \quad (3)$$

The function Ψ' governs the marginal rate of transformation between goods in the morning and goods in the afternoon. For future reference, I record its essential properties here.

Assumption A1. Let Ψ be twice differentiable on $\mathbb{R} \setminus \{0\}$, satisfying $\Psi(0) = 0$, and

$$\begin{aligned} \Psi'(x) &= 1, \Psi''(x) = 0 & x \leq 0, \\ \Psi'(x) &> 1, \Psi''(x) > 0 & x > 0 \end{aligned}$$

When liquidity is abundant, and there is no early liquidation, transformation is neutral and goods are locally homogeneous across sub-periods. When early liquidation is positive, it incurs a convex cost, and liquidity has value.

Taken together, the bank chooses bond holdings b_t and liquidation x_t , taking the price p_t and the support it receives ℓ_t as given,¹ to solve

$$\begin{aligned} \max_{\{b_t, x_t\}} \quad & \mathbb{E}_0 \sum_t \beta^t \Pi_t \\ \text{s.t.} \quad & \lambda_t d_t + p_t (b_t - (1 - \mu)b_{t-1}) = \mu b_{t-1} + x_t + \ell_t \\ & \Pi_t + (1 - \lambda_t)R_t^d d_t = Ad_t - \Psi(x_t) - \ell_t. \end{aligned} \quad (4)$$

¹Treating support as given is without loss of generality. In the absence of liquidity needs, the marginal rate of transformation between goods is 1, so ℓ_t is neutral. With liquidity needs, the rate exceeds 1, and ℓ_t is strictly beneficial for the bank.

2.1.3 First-Order Conditions

The bank's pricing kernel respects

$$p_t = \beta \mathbb{E}_t \left[\frac{\Psi'(x_{t+1})}{\Psi'(x_t)} (\mu + (1 - \mu)p_{t+1}) \right], \quad (5)$$

derived in Appendix B. Positive liquidity needs induce a wedge in the stochastic discount factor of $\Psi'(x_{t+1})/\Psi'(x_t)$. For $x_t, x_{t+1} \leq 0$ the standard case is recovered. Otherwise, since $\Psi''(x_t) > 0$ and $\Psi'(x_t)$ appears in the denominator, scarce liquidity today lowers prices: liquidity-constrained banks look to unwind their buffer stocks, putting downward pressure on the price. Expected liquidity needs tomorrow, meanwhile, push prices up. Policy support ℓ_t acts accordingly. Support for the market today boosts prices, expected support for tomorrow's market depresses them.

A useful benchmark for Ψ is given by an exponential functional form.

Example 1 (Exponential liquidity cost). Early liquidation cost is exponential with parameter α when

$$\Psi'(x) = \exp(\alpha \max\{x, 0\}),$$

giving rise to the pricing wedge

$$\frac{\Psi'(x_{t+1})}{\Psi'(x_t)} = \exp(\alpha \max\{x_{t+1}, 0\} - \alpha \max\{x_t, 0\}).$$

2.1.4 Timing

Having stated the problems of Foreign banks and households, it is useful to state the timing convention explicitly.

1. On entry, households receive $\delta_{i,t}$ and choose deposits $d_{i,t}$, which banks invest.
2. At t_- , liquidity shocks are realized, bonds mature, Domestic chooses b_t and ℓ_t jointly, the bond market clears at p_t , and banks meet withdrawals through sales, support, and liquidation x_t .

3. At t_+ , investment returns are realized, deposits and loans are repaid, and profits are rebated.

Deposits are chosen before the liquidity shock, so banks cannot avoid liquidity risk ex post; policies are chosen afterwards, and incorporate the realization of the stochastic state.

2.2 Issuance Problem

Domestic households order consumption sequences according to

$$U(\{c_t^*\}_{t=0}^\infty) := \mathbb{E}_0 \sum_{t=0}^\infty \beta^t u(c_t^*) \quad (6)$$

where u is strictly concave.

They receive an exogenous stochastic endowment $\{y_t^*\}_{t=0}^\infty$ and are subject to the sequence of budget constraints

$$c_t^* = y_t^* + p_t b_t - (1 - \mu)p_t b_{t-1} - \mu b_{t-1} \quad \forall t. \quad (7)$$

2.2.1 Implementation

A policy consists of issuance decisions $\{b_t\}_{t=0}^\infty$ as well as within-period liquidity support. In the case of practical interest, the latter acts as an effective ceiling on the shadow price of dollar liquidity by putting an upper bound on funding spreads. The natural formulation is as a ceiling on this shadow price. As such, denote by $\bar{x}_t$ a policy which bounds the Foreign marginal rate of transformation above at $\Psi'(\bar{x}_t)$, implemented through the quantity variable ℓ_t .

2.2.2 Domestic Policy

The Domestic policymaker maximizes the utility of the Domestic household subject to the budget constraints, or sequence of Foreign implementability conditions, (7).

The problem depends on two stochastic processes, the Domestic endowment y_t^* and the Foreign liquidity shock λ_t . If both of these are assumed Markov, the policy problem admits a recursive form. A state consists of outstanding liabilities b , the current liquidity shock λ , and the current endowment y^* . Policy functions are issuance $b'(b, \lambda, y^*)$, constrained to be

non-negative; and the ceiling $\bar{x}(b, \lambda, y^*)$. I impose a reduced-form institutional constraint ξ , such that $\bar{x} \geq \xi$. The policymaker cannot compress Foreign liquidation below ξ , so a lower ξ describes a less restricted policymaker and $\xi = \infty$ describes one who never intervenes. This constraint incorporates limits on the capacity for non-standard interventions, and its value indexes the type of the policymaker.

Today's policymaker internalizes the Foreign price schedule generated by (5) conditional on the issuance and support policies of future policymakers. Today's policymaker has no capacity to bind the latter through commitment, and the equilibrium is therefore time-consistent by construction.

Denote the Domestic policymaker's continuation value of a state by W . Its recursive characterization is

$$\begin{aligned} W(b, \lambda, y^*) &= \max_{c^*, b', \bar{x}} \{u(c^*) + \beta \mathbb{E}[W(b', \lambda', y^*)]\} \\ \text{s.t.} \quad c^* &= y^* + p(b', \bar{x}, b, \lambda, y^*) [b' - (1 - \mu)b] - \mu b \\ \bar{x} &\geq \xi, \quad b' \geq 0. \end{aligned} \tag{8}$$

Definition 1 (Markov perfect equilibrium). A Markov perfect equilibrium consists of policy functions $(b', \bar{x})$ and a price function p mapping from states (b, λ, y^*) , and Foreign allocations, such that:

1. Foreign households and banks optimize, with p satisfying (5);
2. $(b', \bar{x})$ maximize W given p , where the expectation in (5) is taken with respect to the law of motion induced by $(b', \bar{x})$;
3. The bond market clears.

3 Swap Lines Flatten Treasury Demand

This section documents the key theoretical implications of the preceding environment.

As an essential input to what follows, I first observe a striking corner property of the optimal time-consistent policy across an important subset of equilibria.

Lemma 1. *If net issuance is positive, $b' > (1 - \mu)b$, and Foreign liquidity is scarce, $x > 0$, then the liquidity ceiling is at its lower bound, $\bar{x} = \xi$.*

Proof: See Appendix A.

Lemma 1 expresses this paper's key dynamic inconsistency. When the policymaker wishes to support demand for its bonds in smoothing Domestic consumption, no interior policy suffices short of full saturation of Foreign liquidity needs. This saturation degrades the liquidity services of assets issued in the previous period. But the policy cannot internalize the resulting erosion of power over a market which has already resolved. A committed plan would trade off consumption smoothing against the erosion of market power. The time-consistent policy trades off consumption smoothing against nothing.

The remainder of the section examines the consequences of an exogenous movement in the ceiling. This exercise is justified by the latter's position at the boundary: movements in the ξ constraint pass through directly to the endogenous policy in cases of interest, and can meaningfully be subject to exogenous perturbations. I illustrate the progression from Lemma 1 to a decline in the convenience yield by discussing:

1. Why demand for liquid assets is 'inelastic', inducing market power;
2. Why a lower liquidity ceiling $\bar{x}$ makes demand *more* elastic; and
3. How this more-elastic demand undermines market power and lowers the premium.

Notation

To reduce notation I suppress dependence on y_t^* and impose the assumption of one-period bonds, $\mu = 1$. This assumption is qualitatively dispensable for all results.

Throughout this section I adopt recursive notation. Denote a price-taking bank's liquidity need

$$x(b, \lambda) = \lambda + p b'(b, \lambda) - b, \quad (9)$$

and the induced survival function of that need,

$$H(z; b) = \mathbb{P}(x(b, \lambda) > z). \quad (10)$$

Define the convenience yield

$$\pi := \beta \left(\frac{1}{\beta} - \frac{1}{p} \right) \equiv \frac{p - \beta}{p}. \quad (11)$$

β is the fundamental price of the asset stripped of its liquidity benefit, under $\mu = 1$. The first definition frames π as a discounted yield spread, following its financial measurement; the latter frames it as a markup, or Lerner index, following the concerns of this analysis.

Much of this section addresses pricing at a ‘non-crisis’ state, consistent with the empirical strategy of section 4. Define such a state as one with strictly negative liquidity requirements, $x(b, \lambda) < 0$ implying $\Psi'(x) = 1, \Psi''(x) = 0$.

3.1 Demand for b' is downward-sloping

The policy problem of section 2 is stated in primal form. Since inverse demand is strictly decreasing, it admits an equivalent dual formulation in which a policy is a price function. The results of this subsection and the next describe a price-taking bank’s response to a perturbation of such a price policy.

Consider a non-crisis state with ceiling $\bar{x}$. Pricing respects

$$p = \beta \int \Psi'(\min\{x(b', \lambda'), \bar{x}\}) dF \quad (12)$$

where F is the distribution over λ' shocks and the denominator is 1 given non-positive current liquidity needs — the ‘non-crisis’ assumption $\Psi'(x) = 1$ for $x < 0$.

This pricing equation admits a more tractable representation in H ,

$$p = \beta \left[1 + \int_0^{\bar{x}} \Psi''(z) H(z; b') dz \right]. \quad (13)$$

Consider the problem of the representative bank, responding to a perturbation in today’s price p taking the future price policy as given. Asset demand is inelastic under transparent conditions.

Proposition 1 (Price elasticity of demand). *The bank's price elasticity of demand respects*

$$\epsilon(b'; \bar{x}) := -\frac{\partial \log b'}{\partial \log p} = \frac{1 + \int_0^{\bar{x}} \Psi''(z) H(z; b') dz}{b' \int_0^{\bar{x}} \Psi''(z) (-H_b(z; b')) dz}.$$

Proof: See Appendix A.

The numerator records the average value of an asset which pays out at t_- : its t_+ value, 1, and its expected relaxation of potential liquidity needs. The denominator records the slope of the contribution.

3.2 Swap lines make demand more elastic

Having established when demand is inelastic, I now consider the consequences of an exogenous downward movement in the liquidity price ceiling $\bar{x}$.

As is clear from the denominator in Proposition 1,

Corollary 1. *For any b' ,*

$$\lim_{\bar{x} \rightarrow 0} \epsilon(b'; \bar{x}) = \infty.$$

This limit isolates the source of inelastic demand. If liquidity needs are always saturated by the policy, we recover exactly the classical benchmark of infinitely elastic demand.

The intuitive directional limit result of Corollary 1 — more liberal support generates more elastic demand — is in fact monotone under mild assumptions.

Proposition 2. *Let $-\partial \log H(z; b') / \partial b'$ be non-decreasing in z . For any $\bar{x}_1 < \bar{x}_2$ the price elasticity of demand is increasing pointwise:*

$$\epsilon(b'; \bar{x}_1) > \epsilon(b'; \bar{x}_2).$$

Proof: See Appendix A.

The intuition is captured in Figure 1, which specializes to the exponential Ψ of Example 1 with $\alpha = 1$. I use that case to obtain $\Psi' = \Psi''$ on $x \in (0, \infty)$, so a single curve depicts both the marginal rate of transformation and its slope. This permits the level and slope of the liquidity premium to be compared as areas in one diagram. Here x_{unc} denotes liquidity needs under no support and Ψ'_{unc} denotes the corresponding unconstrained marginal-cost schedule.

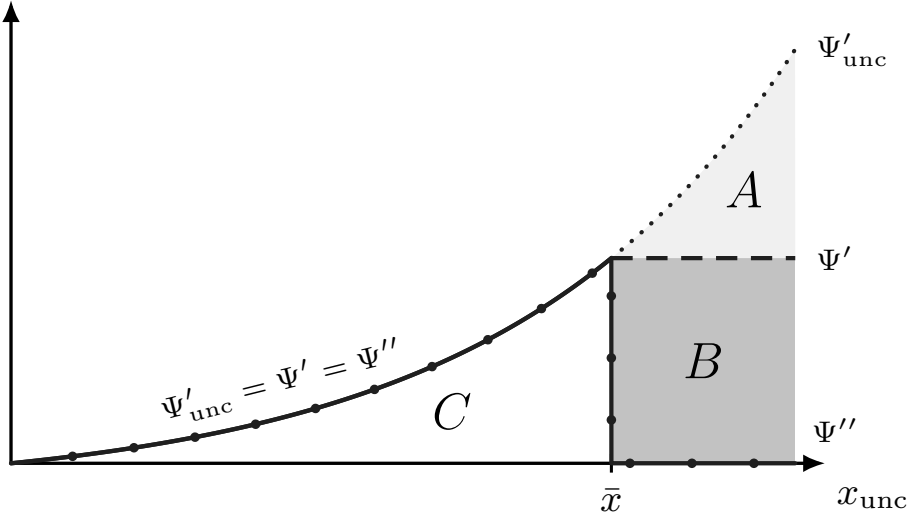


Figure 1: Swap lines flatten the demand slope

Notes: Exponential Ψ from Example 1 with $\alpha = 1$, so $\Psi' = \Psi''$ conditional on $x > 0$. Unconstrained liquidity needs x_{unc} are uniform, making expectations proportional to areas. Beyond $\bar{x}$, Ψ'_{unc} follows the dotted path while realized Ψ' flattens and $\Psi'' = 0$. The premium level retains $B + C$, while its slope retains C , yielding $\epsilon = (B + C)/(b'C)$.

Suppose liquidity needs be uniform over the unconstrained range x_{unc} on the horizontal axis, so that expectations are proportional to areas.²

Up to the ceiling the realized and unconstrained schedules coincide. Beyond it, the unconstrained schedule Ψ'_{unc} continues along the dotted path, while the realized marginal rate of transformation is flat at $\Psi'(\bar{x})$; its slope Ψ'' , meanwhile, collapses to zero. The level therefore retains C and B and loses only A ; the slope retains C alone, losing A and B together. The elasticity is exactly

$$\epsilon(b'; \bar{x}) = \frac{1}{b'} \frac{C + B}{C}. \quad (14)$$

Lowering the ceiling raises B/C , and flattens demand; as $\bar{x} \rightarrow 0$ the slope vanishes and the elasticity diverges, which is Corollary 1.

²The uniform case suppresses the probability weighting. In general the level weights each layer by H and the slope by $-H_b$, and the hazard condition of Proposition 2 rules out cases where the latter dominate at the margin. This assumption provides a formal counterpart to the ‘good friend’ analogy characterizing safe assets (Brunnermeier et al., 2024). Like a good friend, a safe asset is there when you need it. Specifically, a safe asset should not become increasingly powerless when needed most.

3.3 More elastic demand erodes market power

It is intuitive to expect from Proposition 2 that lower $\bar{x}$ will erode the planner's market power more generally at the optimal policy. Theoretical support for this inference becomes technical in the general dynamic setting. Here I extend Proposition 2 to the equilibrium allocation in a static example.

Consider a two-period planner choosing issuance to solve

$$\begin{aligned} \max_{b'} \quad & c_0 + \beta c_1 \\ \text{s.t.} \quad & c_0 = y + p(b'; \bar{x}) b' \\ & c_1 = y - b', \end{aligned} \tag{15}$$

where inverse demand $p(b'; \bar{x})$ is given by (13) with $b'' \equiv 0$, so that $x(b', \lambda') = \lambda' - b'$.

The policymaker's first-order condition is the Lerner condition

$$\pi(b'; \bar{x}) = \frac{1}{\epsilon(b'; \bar{x})}. \tag{16}$$

The pointwise case then extends naturally to equilibrium.

Proposition 3. *Let the shock distribution F have non-decreasing hazard. Then lowering the cap $\bar{x}$ increases the elasticity, increases issuance, and lowers the premium at the optimal issuance:*

$$\frac{d\epsilon^*}{d\bar{x}} < 0, \quad \frac{db'^*}{d\bar{x}} \leq 0, \quad \text{and} \quad \frac{d\pi^*}{d\bar{x}} > 0.$$

Proof: See Appendix A.

4 Estimating the Demand Rotation

The results of section 3 lead to a clear empirical prediction. If the swap line policy realizations of the GFC and March 2020 were in part unanticipated, they should make Foreign Treasury demand more elastic in normal times.

This section tests the hypothesis by estimating rotations in demand among swap line recipients relative to non-recipients. Focusing on slopes allows me to difference out level effects altogether, which may be directly sensitive to the policy with ambiguous sign. Focusing on

relative changes allows me to exclude the aggregate market elasticity, itself an outcome of the unknown policy. The focus on non-crisis buffer-stock accumulation gives enough observations to work in the country cross-section. The resulting estimator can be implemented as a parsimonious difference-in-differences in slopes.

4.1 Data

The essential demand-system estimation requires an annual price p_t and country-level holdings data $b_{i,t}$, each aggregated across outstanding notes and bonds. The sample covers 2002–2024.

I construct the price changes using security-level yields and characteristics from the CRSP U.S. Treasury Database, restricted to notes and bonds. I aggregate security-level yields into a single market-portfolio yield weighting by amount outstanding. The market portfolio’s modified duration D_t is aggregated the same way, and averages roughly 5.5 years over the sample. I recover the price change from the market yield by a first-order duration approximation.

For the quantity cross-section, I use the Treasury International Capital (TIC) SHL surveys. Starting in 2002, these surveys provide annual data on holdings of Treasury debt by holder country of residence based primarily on security-level reports from U.S. custodians and issuers.

I include countries by scale of 2002 total holdings after excluding Belgium, Luxembourg, and the Cayman Islands and other known sources of custodial bias. I form a panel of 34 countries evenly split between swap line recipients and non-recipients, including all TIC Major Foreign Holders.

In line with section 3’s focus on out-of-crisis behavior, I exclude 2020 and 2007–2008. Results are robust to extending the exclusion through 2012 to exclude the Eurozone crisis.

4.2 Specification

To exploit cross-country heterogeneity in the data, I consider an environment featuring a continuum of atomistic countries admitting one representative bank per country. A country-bank i in a period t observes policy $\bar{x}_{i,t}$, such that liquidity needs in the subsequent period

are bounded above by $\bar{x}_{i,t}$.³ If an observed swap line episode reveals information about the true policy, this revelation induces a change in $\bar{x}_{i,t}$ thereafter.

I consider two canonical interventions: The Global Financial Crisis (GFC) and the market disruption of March 2020 (COVID). I partition the sample into three windows,

$$\tau \in \{0, \text{GFC}, \text{COVID}\} := \mathcal{T}. \quad (17)$$

Define the between-window country-level change in anticipated policy

$$\Delta \bar{x}_{i,\tau} := \bar{x}_{i,t} - \bar{x}_{i,s}, \quad t \in \tau, \quad s \in \tau - 1. \quad (18)$$

Here $\tau - 1$ denotes the sample window immediately preceding τ .

To represent secular growth in asset demand, relax the $\delta_{i,t} = 1$ assumption of the theory, and assume instead $\delta_{i,t}$ is a scale shifter such that $b_{i,t}/\delta_{i,t}$ is stationary and $b_{i,t}$ inherits any long-run trend from $\delta_{i,t}$. Define normalized holdings

$$\tilde{b}_{i,t} \equiv \frac{b_{i,t}}{\delta_{i,t}}. \quad (19)$$

The portfolio adjustment is scale invariant, so the bank's problem depends on holdings and scale only through $\tilde{b}_{i,t}$.

Return to the inverse demand curve implied by equation (12). Rewriting as a conditional expectation over future liquidity needs,

$$\begin{aligned} p_t &= \beta \mathbb{E}_t \left[\Psi'_i(x_{t+1}) \middle| \tilde{b}_{i,t}, \bar{x}_{i,t}, Z_{i,t} \right] \\ &:= D_i^{-1} \left(\tilde{b}_{i,t}; \bar{x}_{i,t}, Z_{i,t} \right), \end{aligned} \quad (20)$$

where D_i^{-1} is the inverse demand function in $\tilde{b}_{i,t}$, the existence of which follows from the monotonicity Assumption A1. I allow for arbitrary heterogeneity in Ψ_i and F_i within the confines of that assumption. Within-country variation over time, however, is restricted to

³For simplicity I impose deterministic $\bar{x}_{i,t}$ policies. Of course, agents' priors contain uncertainty. With only one moment of the implied distribution identified, however, the deterministic specification sacrifices little.

come from $\bar{x}_{i,t}$ and a vector of controls $Z_{i,t}$ discussed below.

Inverting and recalling $\tilde{b}_{i,t} = b_{i,t}/\delta_{i,t}$,

$$b_{i,t} = \delta_{i,t} D_i(p_t; \bar{x}_{i,t}, Z_{i,t}). \quad (21)$$

Taking logs and totally differentiating,

$$d \log b_{i,t} = -\epsilon_{i,t} d \log p_t + d \log \delta_{i,t}, \quad (22)$$

where

$$\epsilon_{i,t} \equiv -\frac{\partial \log D(p_t; \bar{x}_{i,t})}{\partial \log p_t} > 0 \quad (23)$$

in line with section 3. Locally,

$$\Delta \log b_{i,t} \approx -\epsilon_{i,t} \Delta \log p_t + \Delta \log \delta_{i,t}. \quad (24)$$

Restrict demand shocks to the form

$$\Delta \log \delta_{i,t} = v_t + \sum_{\tau \in \mathcal{T}} \gamma_i^\tau \mathbf{1}_{t \in \tau} + u_{i,t}. \quad (25)$$

Here v_t is a common aggregate shock and γ_i^τ is country-specific trend demand across a window τ .

Let $SL_i \in \{0, 1\}$ indicate a country i 's inclusion in the realized swap line network. The estimating equation is then

$$\begin{aligned} \Delta \log b_{i,t} &= -\epsilon_{i,t} \Delta \log p_t + \zeta_t + \sum_{\tau \in \mathcal{T}} \gamma_i^\tau \mathbf{1}_{t \in \tau} + \theta'_b Z_{i,t} + u_{i,t} \\ \epsilon_{i,t} &= \bar{\epsilon}_i + SL_i \sum_{\tau \in \mathcal{T} \setminus 0} \rho_\tau \mathbf{1}_{t \in \tau} + \theta'_\epsilon Z_{i,t} \end{aligned} \quad (26)$$

The coefficients ρ_{GFC} and ρ_{COVID} are the objects of interest: they correspond to rotations of the demand curve recording the average effect of swap-line treatment. The elasticity also contains a country fixed effect, to remove pre-treatment heterogeneity across national

demand systems, and controls $Z_{i,t}$ relevant to other correlated sources of demand rotation. The wider demand equation includes time fixed effects ζ_t , country-window fixed effects γ_i^τ , and direct contributions of the controls to demand. The term ζ_t purges common aggregate elasticity movements together with direct aggregate demand shocks v_t .

4.3 What ρ_τ Recovers

If a policy event τ reveals country-level information regarding the true policy, $\Delta\bar{x}_{i,\tau} \neq 0$. A unit decrease in $\bar{x}_{i,t}$ increases the elasticity of demand according to Proposition 2. ρ_τ averages this composite. Consider purely for illustration a benchmark case in which excluded countries anticipated their exclusion. Then

$$\rho_\tau \approx \mathbb{E} \left[\frac{\partial \epsilon_{i,t}}{\partial \bar{x}} \Delta \bar{x}_{i,\tau} \mid SL_i = 1, t > \tau \right]. \quad (27)$$

With $\frac{\partial \epsilon_{i,t}}{\partial \bar{x}}$ negative, less than fully-anticipated swap line support in an episode τ , $\Delta\bar{x}_{i,\tau} < 0$, corresponds to the hypothesis $\rho_\tau > 0$. This effect is derived using only relative changes in elasticities: crucially, the estimator does not rely on (or recover) a common elasticity subject to policy response. It further eschews engagement with underlying secular drivers of safe-asset demand, which remain opaque.

Selection. A confounder must (i) covary with swap line inclusion SL_i and (ii) rotate demand at the episode dates. The second requirement is restrictive. Country-specific demand shocks may load on aggregate factors which also move Treasury prices, and such loadings bias the estimated elasticity. The country $\times \Delta \log p_t$ fixed effect absorbs this bias whenever loadings are stable, so differential exposure across countries is not itself a threat. Identification requires only that the treated-minus-control difference in loadings did not shift at 2008 or 2020.

The most tangible candidate is financial development, which should correlate with both inclusion and the post-GFC changes in regulation and market structure. As a proxy I include in $Z_{i,t}$ each country’s credit-to-GDP ratio from the World Bank’s WDI, interacted with a GFC dummy and entering the elasticity equation directly. Priors dictate attenuation: greater exposure to regulatory stringency should lower, not raise, price sensitivity. I also include predetermined start-of-sample log holdings interacted with each window dummy $\mathbf{1}_\tau$, given

its pronounced predictive power for inclusion and multifaceted correlations with relevant fundamentals.

Initial holdings and financial sophistication do not fully determine inclusion. Institutional and geopolitical alignment with the United States offers a plausibly exogenous residual source of variation after imposing these controls (Aizenman et al., 2022; Sahasrabuddhe, 2019).

A further consideration is endogeneity through failure of the small-open-economy assumption.⁴ I confirm estimates are robust to excluding the largest holders. Bias should again be towards attenuation given trend changes in market shares between groups.

4.4 Results

I estimate four variants of equation (26), first recording only the GFC swap line event, then separating the GFC and March 2020 events. In each case, I consider the model with no additional controls $Z_{i,t}$, and with additional controls for start-of-sample holdings and credit-to-GDP as a proxy for financial development.

Point estimates, standard errors, and hypothesis tests are reported in Table 1. In the GFC-only specifications, which estimate the total effect, the null is convincingly rejected. Swap line recipients became more price-elastic in the post-2008 sample. Adding controls for selection, in particular the financial-development proxy, increases the ρ_{GFC} point estimate from 3.366 to 4.248. While not statistically distinguished, the change may be deemed economically meaningful. Its direction is as anticipated, however: as discussed, more stringent post-GFC regulation among financially developed nations should be expected to lower the elasticity of demand.

Estimating a separate round of treatment post-COVID proves more challenging given a small number of differenced annual observations since the event. The hypothesis tests continue to imply rejection, however, and the point estimates appear coherent: the total effect decomposes into a 1.597 coefficient for the COVID intervention and a 3.510 effect retained by the GFC, with controls. The evidence, together with the theory of section 3, is therefore consistent with the realized swap line policy adjusting expectations and eroding

⁴This effect is mitigated by the small scale of each Foreign holder relative to the *aggregate* (international and domestic) market, which determines the price variation; and the lower volatility of flows among foreign holders relative to domestic (Chaudhary et al., 2026).

U.S. market power.

The post-GFC rotation, $\rho_{\text{GFC}} > 0$, is robust across a range of alternative specifications. The independent COVID contribution is more speculative. In particular, given the small number of periods since COVID, moving to two-way country-year clustering provides a large mechanical reduction in the estimate’s precision, and under some specifications can even lead to a failure to reject at the 10 percent level. Since the appropriate clustering choice is ambiguous, I take the ρ_{COVID} point estimate as a salient empirical moment while treating the test result with caution.

5 From Demand Rotation to Convenience Decline

Section 4 estimates how swap line access rotated recipients’ demand curves relative to non-recipients. Section 3 qualitatively links this increase to a reduction in market power and convenience yields. But the policymaker is not a static monopolist: the recovered demand rotation is a clean empirical moment rather than a sufficient statistic for the markup. I therefore choose post-GFC and post-COVID liquidity ceilings that reproduce the estimated relative rotations, then compute the associated non-crisis convenience yields. The fitted policies reduce the yield from 100 basis points before the GFC to 53 after the GFC and 47 after March 2020.

5.1 Environment Specialization

To clarify the mechanism, I work with a regime-switching framework featuring crisis (C) and non-crisis (NC) states. A regime r switches to its counterpart with independently and identically distributed probability χ_r , $r \in \{C, NC\}$. I adopt the shorthand $\mathbb{E}_C$ ($\mathbb{E}_{NC}$) for “expectation conditional on crisis (non-crisis).”

Periods are quarterly. I impose two functional forms and two distributional assumptions to obtain a quantitative model.

Domestic utility is CRRA with coefficient γ . Ψ is specialized to the exponential case of Example 1,

$$\Psi'(x) = \exp(\alpha \max\{x, 0\}). \quad (28)$$

Table 1: Swap Lines and the Price Elasticity of Treasury Demand

Dependent variable:	$\Delta \log b_{i,t}$			
	GFC only		GFC and COVID	
	(1)	(2)	(3)	(4)
<i>A. Coefficient estimates</i>				
$SL_i \times \text{Post-GFC} \times (-\Delta \log p_t)$	3.366 (1.583)	4.248 (1.626)	2.549 (1.896)	3.510 (1.787)
$SL_i \times \text{Post-COVID} \times (-\Delta \log p_t)$	— —	— —	1.680 (1.032)	1.597 (1.060)
Country fixed effects*	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes
Observations	646	646	646	646
Within R^2	0.139	0.144	0.168	0.174
<i>B. Hypothesis tests: one-sided p-values</i>				
$\rho_{\text{GFC}} > 0$	0.018	0.006	0.076	0.032
$\rho_{\text{COVID}} > 0$	—	—	0.042	0.055

* Country fixed effects include country fixed effects interacted with each regime and a single country $\times \Delta \log p_t$ fixed effect. See section 4.2.

Notes: Panel A reports coefficient estimates with standard errors in parentheses. The parameters ρ_{GFC} and ρ_{COVID} denote the coefficients on the corresponding triple interactions shown in the table.

Additional controls include log start-of-sample bond holdings and WDI credit-to-GDP ratio, independently and interacted with $\Delta \log p_t$ and regime dummies.

Panel B reports wild-cluster-bootstrap one-sided p -values for model predictions shown in the corresponding rows. Each prediction $\rho_\tau > 0$ is evaluated against $H_0 : \rho_\tau \leq 0$.

Standard errors and p -values are clustered at the country level and estimated by wild bootstrap.

λ_t is fixed $\lambda_{NC} \approx 0$ out of crisis and distributed

$$\lambda_t \sim U[\lambda_{NC}, \lambda_{\max}] \quad (29)$$

in crisis. The Domestic endowment is AR(1) with persistence ρ and innovation variance σ^2 . On entering crisis the Domestic endowment suffers a one-time reduction Δy_C then proceeds as before.

I treat the constraint on the liquidity price ceiling, ξ , as exogenous. I model each swap-line expansion as an unanticipated permanent relaxation of the ceiling, so expectations adjust on impact.

The solution algorithm is detailed in Appendix C.

5.2 Pre-GFC Baseline Calibration

For external parameters, I set $\{\gamma, \beta, \rho, \sigma\}$ to literature-consistent values. μ is chosen for expected maturity of 5.5 years. The probability of crisis and recovery χ_C and χ_{NC} induce a crisis of two-year average length, on average every ten years.

Liquidation cost parameter α and maximum liquidity shock $\lambda_{\max}$ jointly target average Foreign-owned debt to U.S. GDP of 0.4; and an annualized convenience yield of 1 percent, consistent with literature estimates (Jiang et al., 2021; Koijen and Yogo, 2026) after excluding crisis events. This yield is defined as the annualized spread between the model asset's yield and the model-implied yield of an illiquid asset.

I normalize the pre-GFC intervention constraint to be non-binding.⁵

The baseline elasticity ϵ_0 is left as an untargeted validation of the implied demand system. At 1.64, it is comfortably in the literature range.

5.3 Recovering the Decline in Convenience Yields

I use the estimated demand rotations to identify the change in anticipated swap-line policy, then compute the implied convenience yields. By Lemma 1, the institutional constraint ξ pins down the equilibrium liquidity ceiling $\bar{x}$ whenever positive net issuance coincides with

⁵This normalization does not introduce an additional assumption. As liquidity shocks have compact support, lowering initial ξ is equivalent to lowering $\lambda_{\max}$.

Table 2: Calibration and Targeted Moments

Parameter	Description	Value	Source / Target
β	Discount factor	0.99	Standard
γ	Risk aversion	2.0	Standard
σ	Endowment s.d.	0.01	Standard
ρ	Endowment persistence	0.95	Standard
μ	Bond maturation	0.045	Maturity 5.5 years
χ_{NC}, χ_C	Regime-change probabilities	0.025, 0.125	Bianchi et al. (2025)
Δy_C	Crisis endowment loss	0.05	Assumption
α	Liquidity cost shifter	2.03	Target: $\mathbb{E}_{NC}[\pi_0]$
$\lambda_{\max}$	Maximum liquidity shock	0.20	Target: $\mathbb{E}_{NC}\left[\frac{b_0}{y^*}\right]$

Targeted Moments

Moment	Description	Value
$\mathbb{E}_{NC}[\pi_0]$	Pre-GFC convenience yield	100 bps
$\mathbb{E}_{NC}\left[\frac{b_0}{y^*}\right]$	Pre-GFC external debt-to-GDP	0.4

scarce Foreign liquidity. I impose this equivalence and verify that the required conditions hold in the ergodic distribution. The pre-GFC ceiling is nonbinding, $\bar{x}_0 = \infty$. I model the GFC and COVID expansions as unanticipated shifts to new ceilings $\bar{x}_{\text{GFC}}$ and $\bar{x}_{\text{COVID}}$.

For each candidate ceiling, I construct the model counterpart of the difference-in-differences coefficient estimated in Section 4. Let $\epsilon(\bar{x})$ denote swap line recipients' mean non-crisis demand elasticity under aggregate policy $\bar{x}$. To recover the corresponding elasticity among non-recipients, I introduce an atomistic bank that faces the same prices and aggregate shocks but receives no liquidity support. Because this bank is of measure zero, its portfolio choice does not affect equilibrium prices. Its idiosyncratic ceiling remains at $\bar{x}_i = \infty$ inducing distinct precautionary demand $b_{i,t}$. Denote its elasticity by $\epsilon_{\infty}(\bar{x})$.

The model-implied relative rotation is therefore

$$\rho_{\text{Sim}}(\bar{x}) \equiv \epsilon(\bar{x}) - \epsilon_{\infty}(\bar{x}). \quad (30)$$

Under the pre-GFC policy the two groups are identical, so $\rho_{\text{Sim}}(\infty) = 0$.

Table 3: Results — Non-Crisis Model Moments

Regime	Rotation target	Fitted ceiling	Convenience yield
Pre-GFC	0.00	∞	100 bps
Post-GFC	3.51	0.06	53 bps
Post-COVID	5.07	0.04	47 bps

I choose $\bar{x}_{\text{GFC}}$ to match the estimated GFC rotation of 3.51. I then choose $\bar{x}_{\text{COVID}}$ to match the cumulative post-COVID rotation of 5.07, equal to the GFC estimate plus the additional COVID estimate. The convenience yields associated with these ceilings are equilibrium outcomes rather than additional targets.

The fitted policies imply a large decline in the convenience yield. The GFC expansion accounts for most of the decline. Matching the estimated GFC rotation of 3.51 requires a ceiling of $\bar{x}_{\text{GFC}} = 0.06$ and lowers the convenience yield by 47 basis points, from 100 to 53. Matching the cumulative post-COVID rotation of 5.07 requires $\bar{x}_{\text{COVID}} = 0.04$ and lowers the yield by another 6 basis points, to 47.

Figure 2 graphs this asymmetry. The mapping from demand rotations to convenience yields is highly nonlinear: the GFC policy crosses the steep portion of the curve, whereas the COVID policy moves along its flatter tail. In policy terms, the fitted ceiling permits support in roughly 70 percent of crisis realizations after the GFC and 80 percent after COVID. The first intervention therefore changed beliefs about crisis support sharply; the second broadened a policy that markets already expected. Notably, the lower precision of the COVID rotation appears comparatively unimportant for the quantitative outcome.

This nonlinearity is also the exercise’s central qualification. The estimated rotations do not identify the yield decline without additional structure, and the result depends on where the calibrated economy lies on the curve.

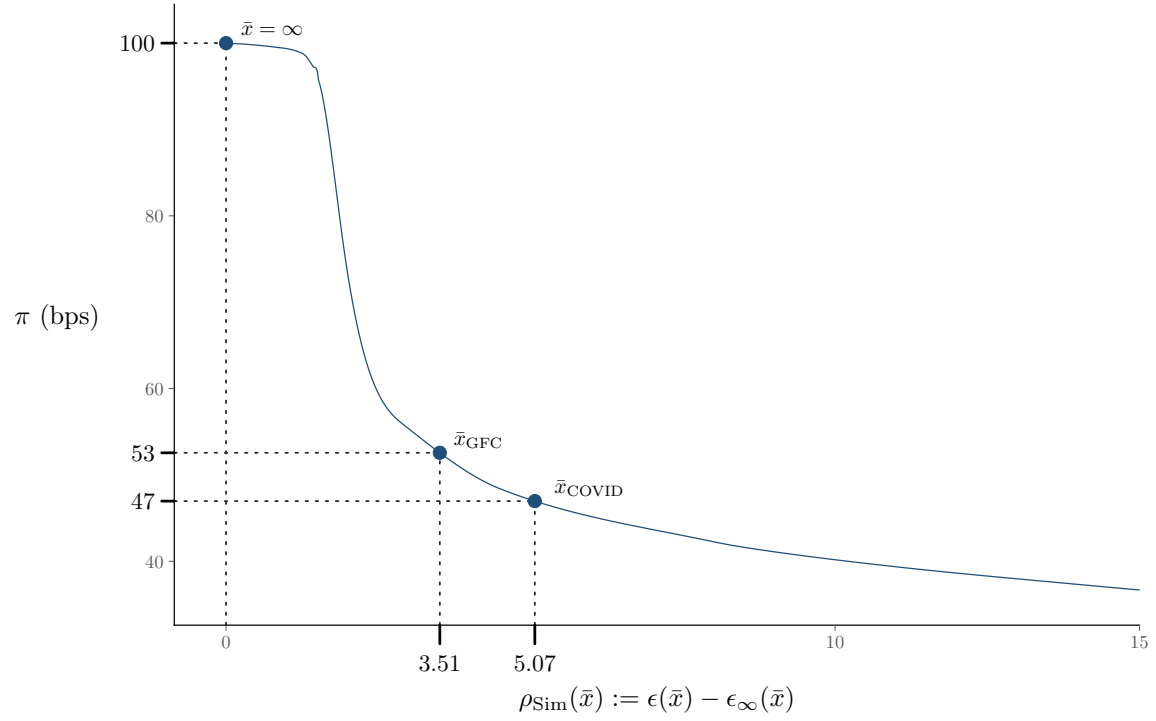


Figure 2: Demand rotations reduce convenience yields

Notes: π is the annualized spread between liquid bond b and the model's illiquid benchmark. ρ_{Sim} is the model-implied elasticity rotation for recipients relative to non-recipients. The curve varies the liquidity ceiling $\bar{x}$ from the never-binding case; $\bar{x}_{\text{GFC}}$ and $\bar{x}_{\text{COVID}}$ match the rotation targets from Table 1, specification 4.

5.4 Crisis Dynamics

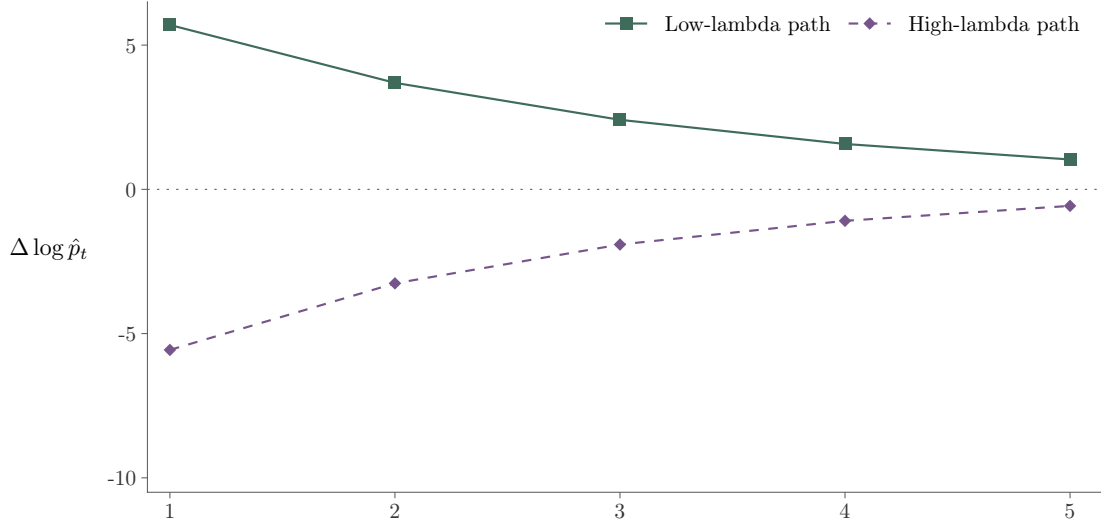
The quantitative exercise targets non-crisis moments, but these are generated by the crisis mechanism. The latter must cohere with observed crisis behavior of Treasury yields.

Consider first a no-support policy, $\bar{x} = \infty$. A crisis moves the Treasury price through three forces. The fall in Domestic endowment Δy_C raises desired issuance as the policymaker smooths consumption. Current dollar needs induce Foreign banks to sell Treasuries, reducing demand. Expected future needs instead raise precautionary demand. The price response depends on the balance between current liquidation and future precautionary demand.

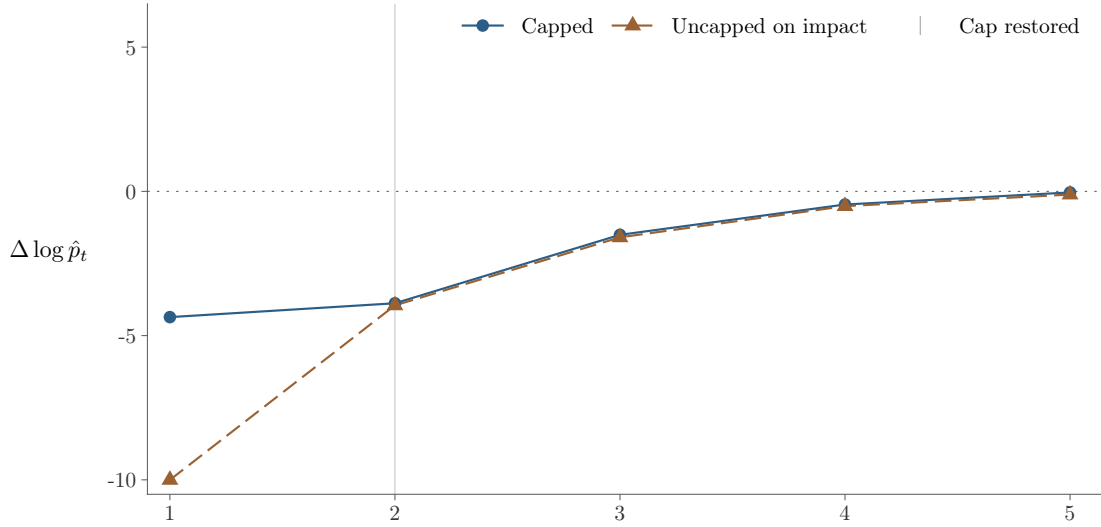
Figure 3(a) illustrates the two possible outcomes. When current liquidity needs are low relative to those expected in subsequent periods, precautionary demand dominates: the Treasury price rises despite greater issuance, as in late 2008. When current needs are sufficiently severe, immediate liquidation dominates and the price falls, as in March 2020. The model therefore reconciles opposite crisis price movements through differences in the timing, rather than simply the presence, of dollar liquidity needs.

Crisis also creates the incentive to intervene. Figure 3(b) holds fixed the path of liquidity shocks and compares the post-GFC ceiling $\bar{x}_{\text{GFC}}$ with a one-shot deviation to no support. Removing the ceiling on impact amplifies Foreign liquidation and more than doubles the increase in U.S. borrowing costs. The Treasury market consequently becomes least effective as a tool for consumption smoothing precisely when the Domestic endowment is lowest.

This experiment provides the quantitative counterpart to Lemma 1. Once a crisis occurs, swap lines support Treasury demand and lower the cost of current issuance. Before the crisis, however, the expectation of that support weakens precautionary demand and reduces the convenience yield. The same intervention that stabilizes the Treasury market ex post therefore erodes U.S. market power ex ante.



(a) Crisis price across liquidity states



(b) Effect of swap line with scarce liquidity

Figure 3: Bond price crisis impulse response functions

Notes: Both panels begin when a crisis arrives from the non-crisis fixed point, $y^* = 1$. $\Delta \log \hat{p}_t$ denotes $100 \times \log$ deviation from the fixed point. Panel (a) sets λ_1 to λ_{NC} (low) or $\lambda_{\max}$ (high), then uses $\lambda_{t+1} = w\lambda_t + (1-w)\mathbb{E}_C[\lambda]$. Panel (b) sets $\lambda_1 = \lambda_{\max} + \Delta$, then reverts to $\mathbb{E}_C[\lambda]$. The capped path uses $\bar{x} = 0.06$; the uncapped path removes the cap only in period 1 before optimal policy resumes.

6 Conclusion

This paper advances a theoretically plausible and quantitatively powerful explanation of falling post-crisis convenience yields on U.S. public debt. The mechanism presents a rational policy under genuine constraints, in contrast to an account of simple mismanagement. It further draws on a view of crisis which incorporates the observed equivocal effects of liquidity shortfall on the bulk of the Treasury market.

The perspective currently remains at the level of conjecture, for three reasons.

1. The estimated rotations in demand are vulnerable to selection. Projection of the treatment onto an identified source of admissible variation, such as geopolitical alignment, would be more compelling than simply controlling for certain observables.
2. The nonlinear map from those rotations to the fall in the premium is underdetermined. The higher-order moments of the λ distribution matter, and should be empirically grounded. This discipline could be drawn from empirical proxies of international dollar liquidity needs, such as CIP deviations.
3. The parsimonious framing emphasized throughout must be shown to cohere with the wider macroeconomic facts to be convincing. Such coherence would require a quantitatively grounded elaboration of the brief discussion in [section 5.4](#).

Each of these channels offers an exciting direction for future work.

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A Proofs

Throughout this appendix b denotes the issuance argument of H , corresponding to b' in the main text. Denote

$$\mathcal{A}(b; \bar{x}) = 1 + \int_0^{\bar{x}} \Psi''(z) H(z; b) dz, \quad \mathcal{B}(b; \bar{x}) = - \int_0^{\bar{x}} \Psi''(z) H_b(z; b) dz,$$

so that $p = \beta \mathcal{A}$ by (13) and $\epsilon = \mathcal{A}/(b\mathcal{B})$.

Proof of Lemma 1

Let the constraint $\bar{x} \geq \xi$ have Lagrange multiplier $\eta \geq 0$. Since $\bar{x}$ is not a state, the continuation value is unaffected by it, and the first-order condition in $\bar{x}$ is simply

$$u_c(c^*) \frac{\partial p}{\partial \bar{x}} [b' - (1 - \mu)b] + \eta = 0.$$

Where the ceiling binds, $\partial x / \partial \bar{x} = 1$, so $\partial p / \partial \bar{x} < 0$ whenever $x > 0$, by Assumption A1 and the kernel (5). With $b' > (1 - \mu)b$ the first term is strictly negative, forcing $\eta > 0$. Where the ceiling is slack, $\partial p / \partial \bar{x} = 0$, but then lowering $\bar{x}$ below $x > 0$ raises p strictly and is feasible whenever $\bar{x} > \xi$. In either case $\bar{x} = \xi$. $\square$

Proof of Proposition 1

For any $x \geq 0$, continuity of Ψ' and the normalization $\Psi'(0) = 1$ give

$$\Psi'(\min\{x, \bar{x}\}) = 1 + \int_0^{\bar{x}} \Psi''(z) \mathbf{1}\{x > z\} dz.$$

Evaluating at $x(b, \lambda)$, taking expectations, and switching the order of integration gives (13) from (12), since $\mathbb{E}[\mathbf{1}\{x(b, \lambda) > z\}] = H(z; b)$ by definition.

Differentiating through with respect to b ,

$$\frac{\partial p}{\partial b} = \beta \int_0^{\bar{x}} \Psi''(z) H_b(z; b) dz = -\beta \mathcal{B} < 0.$$

$p(\cdot; \bar{x})$ is strictly decreasing, and therefore invertible, defining inverse demand function

$b(p; \bar{x})$. And then $\partial b / \partial p = -1/(\beta \mathcal{B})$ (inverse function theorem). Multiplying by $p/b = \beta \mathcal{A}/b$ and multiplying by -1 gives $\epsilon = \mathcal{A}/(b\mathcal{B})$, which is the version in the main text. $\square$

Proof of Proposition 2

Fix b . $\epsilon = \mathcal{A}/(b\mathcal{B})$, so it is sufficient to show $\mathcal{B}/\mathcal{A}$ is strictly increasing in $\bar{x}$. We have $\mathcal{A}'(\bar{x}) = \Psi''(\bar{x})H(\bar{x})$ and $\mathcal{B}'(\bar{x}) = -\Psi''(\bar{x})H_b(\bar{x})$, so we can write

$$\frac{d}{d\bar{x}} \left[\frac{\mathcal{B}}{\mathcal{A}} \right] = \frac{\Psi''(\bar{x})H(\bar{x}) \left[\frac{-H_b(\bar{x})}{H(\bar{x})} \mathcal{A}(\bar{x}) - \mathcal{B}(\bar{x}) \right]}{\mathcal{A}(\bar{x})^2}.$$

Since $-H_b/H$ is non-decreasing by assumption, $-H_b(z)/H(z) \leq -H_b(\bar{x})/H(\bar{x})$ for $z \leq \bar{x}$, so

$$\mathcal{B}(\bar{x}) = \int_0^{\bar{x}} \Psi''(z)H(z) \frac{-H_b(z)}{H(z)} dz \leq \frac{-H_b(\bar{x})}{H(\bar{x})} [\mathcal{A}(\bar{x}) - 1] < \frac{-H_b(\bar{x})}{H(\bar{x})} \mathcal{A}(\bar{x}),$$

where the last inequality is strict because $H_b(\bar{x}) < 0$ and $\Psi'(\bar{x}) > \Psi'(0) = 1$. The term in brackets is therefore strictly positive, implying the result. $\square$

Proof of Proposition 3

With no next-period policy $x(b, \lambda) = \lambda - b$, so $H(z; b) = \mathbb{P}(\lambda > z + b)$ depends on (z, b) only through $z+b$ and $-H_b(z; b)/H(z; b)$ reduces to the hazard of F ; denote this h_F . Proposition 2 then applies under the non-decreasing F hazard assumption.

Differentiating (13),

$$p_b = -\beta \mathcal{B} < 0, \quad p_{\bar{x}} = \beta \Psi''(\bar{x})H(\bar{x}; b) > 0, \quad \text{and} \quad \mathcal{B}_{\bar{x}} = -\Psi''(\bar{x})H_b(\bar{x}; b) \geq 0.$$

Write marginal revenue $\text{MR}(b, \bar{x}) \equiv p + p_b b = \beta[\mathcal{A} - b\mathcal{B}]$. The first-order condition is $\text{MR} = \beta$, that is

$$\int_0^{\bar{x}} \Psi''(z) [H(z; b) + b H_b(z; b)] dz = 0. \quad (\text{A.1})$$

Dividing $\text{MR} = \beta$ by p and using $-p_b b/p = 1/\epsilon$ and $\beta/p = 1 - \pi$ gives $\pi^* = 1/\epsilon^*$, which is standard.

From above,

$$\text{MR}_{\bar{x}} = \beta [\mathcal{A}_{\bar{x}} - b\mathcal{B}_{\bar{x}}] = \beta \Psi''(\bar{x}) [H(\bar{x}; b) + b H_b(\bar{x}; b)].$$

Write $H + bH_b = H[1 - b(-H_b/H)]$. The term in brackets is non-increasing in z by the hazard assumption. If it were strictly positive at $z = \bar{x}$ it would be strictly positive on all of $[0, \bar{x}]$, making the integral in (A.1) strictly positive. Hence $H(\bar{x}; b^*) + b^* H_b(\bar{x}; b^*) \leq 0$ and $\text{MR}_{\bar{x}} \leq 0$. The second-order condition gives $\text{MR}_b < 0$, so differentiating $\text{MR} - \beta = 0$ gives $db^*/d\bar{x} = -\text{MR}_{\bar{x}}/\text{MR}_b \leq 0$, which is strict unless $H + bH_b$ vanishes at $\bar{x}$.

Then $dp^*/d\bar{x} = p_{\bar{x}} + p_b db^*/d\bar{x} > 0$. Since $\pi = 1 - \beta/p$ is strictly increasing in p , $d\pi^*/d\bar{x} > 0$.

Then from above $\epsilon^* = 1/\pi^*$, so $d\epsilon^*/d\bar{x} = -(\pi^*)^{-2} d\pi^*/d\bar{x} < 0$. □

B Bank's Problem

This appendix collects the supporting results for section 2: the deposit corner and the irrelevance of the deposit rate; the irrelevance of scale; the derivation of the pricing kernel; and the implementation of the liquidity ceiling through the quantity of support.

B.1 Deposits and the Deposit Rate

Consider household i at the beginning of period t , before its type is known. Depositing $d_{i,t}$ and storing the remainder, it consumes $\delta_{i,t}$ at t_- if impatient — deposits are redeemable at par in the morning, and undeposited endowment is already a morning good — and $R_t^d d_{i,t} + s(\delta_{i,t} - d_{i,t})$ at t_+ if patient. Expected consumption is therefore

$$c_{i,t} = \lambda_t \delta_{i,t} + (1 - \lambda_t) [R_t^d d_{i,t} + s(\delta_{i,t} - d_{i,t})] + \Pi_{i,t}, \quad (\text{B.1})$$

which is increasing in $d_{i,t}$ whenever $R_t^d \geq s$. An impatient household is indifferent between depositing and storing; a patient household strictly prefers to deposit. Deposits are weakly dominant and $d_{i,t} = \delta_{i,t}$. Evaluated at the corner, (B.1) is the household objective (1) of the main text.

The deposit rate is not pinned down within the range consistent with full deposit, $R_t^d \geq$

s. It is also immaterial. Substituting the bank's afternoon constraint (3) into (B.1) and aggregating, Foreign consumption is

$$c_t = \lambda_t d_t + A d_t - \Psi(x_t) - \ell_t, \quad (\text{B.2})$$

in which R_t^d does not appear: because households own the banks, the deposit rate is a transfer within the Foreign country. Nor does R_t^d enter the morning budget constraint (2), where the bank's obligation to withdrawing depositors is the principal $\lambda_t d_t$. It therefore affects neither liquidity needs nor any equilibrium price, and the normalization $R_t^d = 1$ of the main text is without loss.

It remains to check that banks are willing to absorb these deposits. Holding bond positions and support fixed, $\partial x_t / \partial d_t = \lambda_t$ from (2), so marginal profit on a deposit is $A - (1 - \lambda_t) - \lambda_t \Psi'(x_t)$. Since $\Psi' \geq 1$ by Assumption A1, the subtracted terms are a convex combination of 1 and $\Psi'(x_t)$, and the following is sufficient.

Assumption A2. The state space is bounded, so that outstanding liabilities and liquidity needs take values in compact sets, and $A \geq \Psi'(x^{\max})$, where $x^{\max}$ denotes the largest liquidity need attainable.

Under Assumption A2 the corner $d_{i,t} = \delta_{i,t}$ is an equilibrium, and not merely a household best response.

B.2 Scale

Banks are identical and face the common shock λ_t , and the bank's problem (4) depends on the household sector only through aggregate deposits

$$d_t = \int_0^1 d_{i,t} di = \int_0^1 \delta_{i,t} di =: \bar{\delta}. \quad (\text{B.3})$$

Dispersion in endowments across households is therefore irrelevant: only the mean enters. Let scale be constant over time, and let the liquidation technology be homogeneous of degree one in it, so that a bank of size δ liquidating x bears cost $\delta \Psi(x/\delta)$ and marginal cost depends on liquidation per unit of scale alone.

Dividing the morning and afternoon constraints (2)–(3) by $\bar{\delta}$, and writing hats for per-unit-of-scale quantities,

$$\lambda_t + p_t \left(\hat{b}_t - (1 - \mu)\hat{b}_{t-1} \right) = \mu\hat{b}_{t-1} + \hat{x}_t + \hat{\ell}_t, \quad (\text{B.4})$$

and Ψ' is homogeneous of degree zero, so it depends on $\hat{x}_t$ alone. The pricing kernel (5) is accordingly unchanged, and the price schedule facing Domestic depends on issuance only through $\hat{b}_t$. Setting $\bar{\delta} = 1$ is thus a choice of units for bond quantities, and is the normalization imposed in the main text. Domestic quantities are measured in the same units; the full equilibrium is invariant to the choice provided u is homothetic, which holds in the risk-neutral case of Proposition 3 and under the CRRA specification of section 5.

B.3 The Pricing Kernel

Substituting the afternoon constraint (3) into the objective and attaching the multiplier $\beta^t \theta_t$ to the morning constraint (2), the Lagrangian for (4) is

$$\mathcal{L} = \mathbb{E}_0 \sum_t \beta^t \left\{ A d_t - \Psi(x_t) - \ell_t - (1 - \lambda_t) R_t^d d_t + \theta_t [\mu b_{t-1} + x_t + \ell_t - \lambda_t d_t - p_t (b_t - (1 - \mu)b_{t-1})] \right\}. \quad (\text{B.5})$$

The bank takes p_t and ℓ_t as given and chooses b_t and x_t . Differentiating in x_t ,

$$\theta_t = \Psi'(x_t), \quad (\text{B.6})$$

so the multiplier on the morning constraint is the marginal cost of morning goods in terms of afternoon goods. A bond bought in the morning of t delivers μ in matured principal plus $(1 - \mu)p_{t+1}$ in resale value in the morning of $t + 1$, both of which are morning goods and so valued at θ_{t+1} . Differentiating in b_t ,

$$\theta_t p_t = \beta \mathbb{E}_t [\theta_{t+1} (\mu + (1 - \mu)p_{t+1})]. \quad (\text{B.7})$$

Substituting (B.6) gives the kernel (5) of the main text.

B.4 Liquidity Support and the Ceiling

Write the bank's gross liquidity need, before any support, as

$$z_t = \lambda_t d_t + p_t (b_t - (1 - \mu)b_{t-1}) - \mu b_{t-1}, \quad (\text{B.8})$$

so that (2) reads $z_t = x_t + \ell_t$: support and liquidation are the two ways of meeting the same shortfall. Support is repaid at par, while liquidation costs $\Psi'(x_t) \geq 1$ at the margin, so the bank accepts whatever support is offered in full, as claimed. It is strictly better off doing so whenever liquidity needs are positive, and indifferent otherwise.

Support is extended after λ_t is realized, so the policymaker observes (B.8) when choosing ℓ_t . Setting

$$\ell_t = \max\{z_t - \bar{x}_t, 0\} \quad \text{delivers} \quad x_t = \min\{z_t, \bar{x}_t\}, \quad (\text{B.9})$$

and hence $\Psi'(x_t) \leq \Psi'(\bar{x}_t)$: the quantity of support implements a ceiling on the Foreign marginal rate of transformation, which is the formulation of section 2. Since the price depends on support only through the realized liquidation entering (5), the ceiling $\bar{x}_t$ is a sufficient description of policy and the quantity ℓ_t need not be tracked separately.

Banks are competitive and take the support they receive as given, exactly as they take the bond price. A bank contemplating a larger bond position therefore expects to fund it by liquidating at marginal cost $\Psi'(x_t)$, not by drawing additional support, which is what leaves $\Psi'(x_t)$ in the denominator of (5). The ceiling is a property of the equilibrium that the policymaker's lending delivers, rather than a commitment made to any individual bank.

C Numerical Solution

I solve the recursive equilibrium of section 5 by iteration on the Domestic value function, the issuance policy, and the Foreign bond-price schedule. Let the state be (b, λ, y^*) , with the crisis regime included in the transition law for λ , and write the price schedule as $p(b', b, \lambda, y^*)$. Given a fixed liquidity ceiling $\bar{x}$, define liquidation by

$$x(b', b, \lambda; p) = \min \{ \lambda + p [b' - (1 - \mu)b] - \mu b, \bar{x} \}. \quad (\text{C.1})$$

The marginal liquidation cost is normalized to one when liquidity is not scarce, as in Assumption A1.

Write $B(b, \lambda, y^*)$ for the issuance-policy function. I approximate $W(b, \lambda, y^*)$, $B(b, \lambda, y^*)$, and $p(b', b, \lambda, y^*)$ on finite grids. The endowment process is discretized by the Tauchen method.

Starting from guesses (W_0, B_0, p_0) , each outer iteration has three steps. First, taking (W_n, p_n) as given, I update issuance state by state:

$$\begin{aligned} \tilde{B}_{n+1}(b, \lambda, y^*) = \arg \max_{b'} & \left\{ u(y^* - \mu b + p_n(b', b, \lambda, y^*) [b' - (1 - \mu)b]) \right. \\ & \left. + \beta \mathbb{E}[W_n(b', \lambda', y^*) \mid \lambda, y^*] \right\}. \end{aligned} \quad (\text{C.2})$$

Let B_{n+1} denote the damped combination of the candidate update $\tilde{B}_{n+1}$ and its predecessor B_n .

Second, holding the updated issuance rule and current price schedule fixed, I update the value function:

$$\begin{aligned} W_{n+1}(b, \lambda, y^*) = & u(y^* - \mu b + p_n(B_{n+1}, b, \lambda, y^*) [B_{n+1} - (1 - \mu)b]) \\ & + \beta \mathbb{E}[W_n(B_{n+1}, \lambda', y^*) \mid \lambda, y^*] \end{aligned} \quad (\text{C.3})$$

using multiple iterations per loop.

Third, I update the entire price schedule from the Foreign pricing kernel. For each (b', b, λ, y^*) , let

$$b'' = B_{n+1}(b', \lambda', y^*), \quad p' = p_n(b'', b', \lambda', y^*).$$

The undamped price update is

$$\tilde{p}_{n+1}(b', b, \lambda, y^*) = \beta \frac{\mathbb{E}[(\mu + (1 - \mu)p') \Psi'(x(b'', b', \lambda'; p')) \mid \lambda, y^*]}{\Psi'(x(b', b, \lambda; p_n(b', b, \lambda, y^*)))}. \quad (\text{C.4})$$

The expectation uses the joint transition probabilities for the liquidity regime and Domestic endowment, including the one-time endowment loss upon entry into crisis.

The issuance and price updates are then combined under damping with their predecessors.

Throughout solving the model the liquidity ceiling is held fixed at the boundary, $\bar{x} = \xi$. The net issuance hypothesis of Lemma 1 is then verified to hold across the subset of the ergodic distribution where the ceiling binds.

Non-crisis moments are estimated as $\mathbb{E}_{NC}[\cdot]$ using 10,000 post-burn-in draws. The price elasticity at each state is computed by a centered finite difference of the solved price schedule with respect to b' , holding (b, λ, y^*) .